

VTAN 303 (ANIMAL AND DAIRY PRODUCTS PROCESSING)

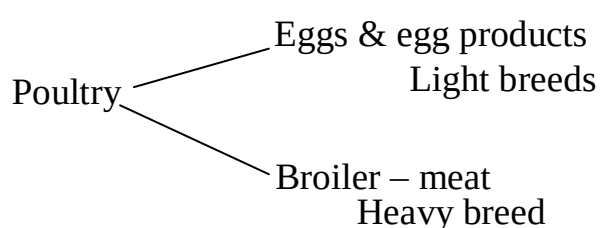
The various species of animals raised by man as sources of high quality animal protein necessary for good health and growth and reproduction.

The quality of protein depends on;

Amount of essential amino acids present

Proportion of the essential amino acids

Digestibility of the essential amino acids



Swine – Pork, Bacon, Lard

Goat & cattle – meat, milk, hides & skin

Sheep – meat, wool

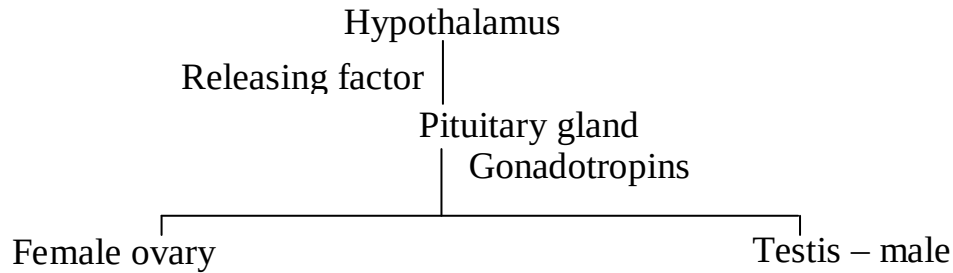
EGG AND EGG PRODUCTS

Special strains of chickens are bred for large scale egg production and are called laying birds.

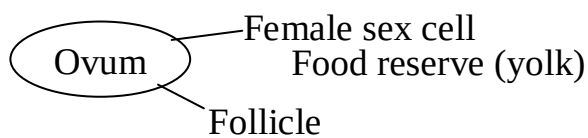
Commercial breeds of laying birds include Harco, shavers and white leghorn. They have been developed for high egg production which can reach 240 – 280 eggs year.

The bulk of eggs produced are consumed in the form of shell egg, some can be frozen (frozen egg), others dried (dried eggs). Most of these are used in the bakery, confectionery and noodle industries. The egg white or albumen is also used in the chemical and pharmaceutical industries.

Eggs are produced and man have decided to be using it for food. Egg serve as a means of reproduction of the bird and the formation of an egg is a function of the hen's reproductive system which consist of the ovary and evident.



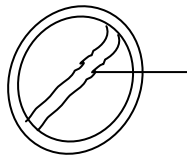
At hatching timer the female chicken have two ovaries and two oviducts. As the chicks continue to grow the right ovary and oviduct usually remain dormant and the left ovary and oviduct develop so as at sexual maturity you have only the left ovary and oviduct. The ovary appears as a cluster of developing ova or yolk each enclosed in a yolk sac of follicle.



The ovary is situated about mid-way along the back bone. The ovary contains about 3600 – 4000 minutes ova (future yolks). When the growing pullet reaches sexual maturity or come into production (20 – 22weeks, some of the ova develop into mature yolks.

“The egg seen is said to develop 10 days ago. When the chicken is about to lay (matured) feed intake increases so as to provide nutrients to the developing egg. Nutrients are transported through blood vessels to the egg”.

The yolk starts as a single cell with vitamin membrane (outer covering) around it. This is enclosed in the yolk sac which contains a vast network of blood vessels. These blood vessels carry yolk building materials/ substances (mainly protein and lipids) to the developing yolk. The deposition of the yolk building materials brings about the growth of the yolk. The yolk begins to grow until it reaches a mature size



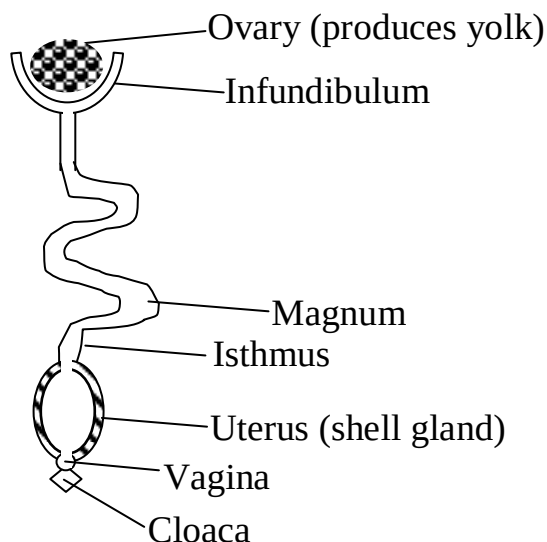
Stigma – an area without blood vessels.

When the stigma breaks, it is known as ovulation.

Ovulation is influenced by LH while growth of the egg is under the influence of FSH.

When the yolk comes to full size, it is released from the ovary by the rupture of the follicle along the stigma. The yolk is discharged into the body cavity and it is engulfed by the upper end of the oviduct known as

INFUNDIBULUM



Female reproductive tract of the chicken which is funnel shaped. The oviduct is a tube-like organ approximately 63 – 68cm long which is divided into five (5) areas, each performing certain function in completing the formation of the egg.

INFUNDIBULUM: This is the funnel shape part of the oviduct which receive the ovulated yolk and is the site of fertilization if the egg is to be fertile. For the egg to be fertile there must be sperm present at the infundibulum.

MAGNUM: It is the longest part of the oviduct and secretes thick white or albumen.

ISTHMUS: This is the organ which adds the two shell membranes

UTERUS (SHELL GLAND): This part secretes the shell which gives the egg its characteristics shape. It secretes shell, shell pigment and fluid.

UTERO-VAGINAL JUNCTION: Consist of specialized cells that store spermatozoa and is why the female can lay fertile eggs for about 10 – 14 days.

VAGINA: Passage way for fully developed eggs into the cloaca.

CLOACA: Is a specialized organ which is a common opening for digestive, reproductive and respiratory purposes.

The vagina is a passage way for developed eggs into the cloaca. Furthermore, the utero-vaginal function have specialized sperm storage cell which help in sperm storage once the hen has mated. Thus a hen that has mated can continue to lay fertile eggs for 10 – 14 days without mating again. The time interval between the ovulation of a yolk from the ovary and the laying of a complete egg varies from about 26-30 hours. The implication of this is that a hen cannot lay more than one egg a day.

The time of lay is influenced by photoperiod it is too late in the evening to lay the hen will retain the egg and lay the egg early the following morning. These chickens lay in clutches (series of laying before skipping a day) and that is why chickens cannot lay everyday throughout the year is 365 eggs/year

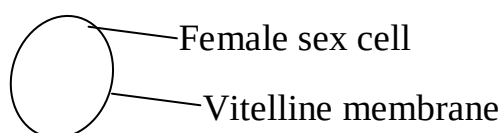
Section of oviduct	Approx length	Approx time
Infundibulum	9cm	15mins
Magnum	33cm	3 hours
Isthmus	10cm	1¼ hours
Uterus	12cm	21-22 hours
Vagina	12cm	Passage way to cloaca

Thus it takes more a day to form an egg and a hen cannot continue to lay throughout the year without skipping a day.

PHYSICAL STRUCTURES OF THE EGG

The avian egg is made up of the yolk, albumen, shell membranes and the shell.

YOLK: The yolk is formed in the ovary. It is usually yellow in colour and the yellow colour is due to the deposition of pigments known as xanthophylls. The yolk is covered by a membrane known as the vitelline membrane. The female sex cell floats on top of the yolk. If the egg is to be fertile the spermatozoa must penetrate the vitelline membrane and fertilize



the female sex cell to form the embryo. The size of the yolk increases with the age of the laying hen.

ALBUMEN OR EGG WHITE: It is formed in the magnum or oviduct and is the largest portion of the egg. It consists of two parts; the thick and thin albumen. The thick albumen is very thick like a gel and it is prominent in fresh eggs. The thin albumen is very watery.

SHELL MEMBRANE: There are two shell membranes in the egg, the inner and outer shell membranes. The shell membranes are formed in the isthmus portion of the oviduct. Between the inner and the outer shell membrane is the air cell, the size of the air cell gives an indication of the age of the egg. Fresh eggs do not have an air cell.

SHELL: The shell is made up of deposit of calcium carbonate. The shell is to protect the egg content the quality of the shell of an egg is very important because the egg is a very perishable product. There are pores inside the shell (for embryo dev). The shell takes a very long time to form and is formed in the uterus or shell gland. Nutrition is very important in determining the quality of the shell (Ca, P, vit D₃ for shell & bone formation). Shell pigments are present in the uterus are responsible for shell colour depending on the breed of the bird.

Interms of the physical structure of the egg, the shell and the shell membranes by weight constitutes about 11% of the egg, the yolk about 30% of the egg and the albumen about 59% of the egg.

THE CHEMICAL COMPOSITION OF THE EGG

Research has shown that, the main factors influencing the proportion of egg components are species, strain, egg size, season. Large eggs contain proportionally more egg white and less yolk than the small egg. It has been shown that as birds advance in age the percentage of yolk per egg is percentage of thick and thin albumen in eggs varies with strain, individual birds, freshness, holding conditions and holding time.

XCAL COMPOSITION

The chemical composition of an egg is influenced by strain, diet, age and season of lay.

WHOLE EGG:

COMPONENT	PERCENTAGE
Water (H ₂ O)	65%
Protein	12%
Fat (lipids)	11%
Ash (mineral)	11%
Carbohydrate (CHO)	1%

The dry (protein & fat) matter of whole egg minus the shell is mainly protein and fats.

ALBUMEN	PERCENTAGE
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COMPONENT	
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Water (H ₂ O)	88%
Protein	10
Fat (lipids)	-
Ash (mineral)	1
Carbohydrate (CHO)	1

The dry matter of the albumen is mainly protein

YOLK	PERCENTAGE
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COMPONENT	
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Water (H ₂ O)	49%
Protein	17%
Fat (lipids)	33%
Ash (mineral)	1%
Carbohydrate (CHO)	-

The dry matter of the yolk is mainly fats and protein and you have more fat than protein.

The embryos grow on the egg which is very nutritious in terms of protein and fats which is a concentrated form (source) of energy (fats is 2.25 times of energy in CHO 9/4).

EGG QUALITY

Egg quality refers to the sum total of the attributes of an egg which make it acceptable to the consumers. Thus, it is related to consumer satisfaction and it includes such factors as appearance colour, odour, flavor and nutritional value. Quality factors of an egg include;

- (1) Uniformity in interior quality – condition of the yolk and albumen.
- (2) Soundness of the shell

- (3) Yolk colour; there must be a colourant or yellow maize to give xanthophylls
- (4) The size and shape of the egg; from marketing and consumer point of view they want large eggs. Egg size can be influenced by protein increase and methionine.
- (5) Appearance

Quality in eggs is influenced by a number of factors. The most important of which can be classified under the following;

1. Inheritance or genetics
2. Management – housing equipment for egg handling, candling, grading, packaging & monitoring
3. Health – certain diseases affect the quality of the egg e.g Newcastle infectious bronchitis leads to mis-shaped eggs.
4. Environment – ambient temperature. If the ambient temperature is very high it reduces quality.
5. Age of birds and nutrition. In general as laying birds age the size of the egg increases and the shell quality decreases.
6. Eggs are delicate and perishable food products that can deteriorate in quality very rapidly under adverse conditions. It is important that eggs reach the consumer with the least possible loss of original quality.

Note that as soon as the egg is laid the quality begins to decline. However, the rate of the decline can be reduced by management practices.

Where you cannot refrigerate like in the rural areas, coat your eggs with oil (glnut) the pores are covered and reduces the exit of CO₂ and water and it can be preserved for some time also by keeping in CO₂ environment.

EGG QUALITY FACTORS

- External quality – colour of the shell, appearance (clean dirty) shape of the egg and egg size
- Internal quality factors – condition of the yolk and egg white. The factors that affect the internal quality include;
 - Meat spots inside the egg
 - Blood spots inside the egg
 - Watery albumen

As the egg ages the thick albumen will be changing from the gel to watery condition. And the reason for this – the increase in PH of the egg as egg ages is responsible for the watery condition of the albumen.

To preserve the internal quality of the egg e need to reduce the rate of CO₂ loss from the egg.

EXTERNAL QUALITY

The two most important external quality are egg size and shell quality. From the stand point of marketing, we prefer large eggs to small eggs, because of this need we try to delay sexual maturity in laying hens. There are methods used to delay sexual maturity

- Restriction of feed intake
- Reducing of photoperiod

Research has shown that egg size increases with the age of the laying hen and delay of sexual maturity increases the size of the egg at the onset of the lay. Eggs are usually classified according to size.

Jumbo	—	Have their shell	
Very large	}	Market eggs should be in this egg range	These depends on the weight of the
Large			
Medium			
Small			
Peewee			

FACTORS AFFECTING EGG SIZE

1. **INHERITANCE OR GENETICS:** Improvement or maintenance of good egg size in a flock maintenance of good egg size in a flock can relatively be accomplished by selective breeding. This is because of high heritability of egg weight.
2. **NUTRITION:** Nutrition has a definite influence on the size of the egg, increase in the feed intake of dietary protein will increase the egg size.

Protein level	Hen day production	Average egg weight
12	60.5	55 grams
14	75.8	57.4
16	74.2	58.5
18	74.1	59.8

Feeding high level of protein at the onset of production will help to increase egg size more rapidly. Apart from the dietary protein, other factors influencing the egg size include;

- Level of methionine or cystine (but cannot be converted to methionine).
 - Linoleic acid level
3. **ENVIRONMENTAL TEMPERATURE:** (When temperatures are high there is reduction in feed intake invariably leading to small eggs). Egg size decreases as temperature increases above 18°C and drops sharply as the air temperature rises above 29°C. because of this in developed countries laying hens are kept in air conditioned rooms.
 4. **AGE OF THE BIRD:** On the average egg size increases with the age of the bird.
 5. **DISEASES:** Certain diseases apart from affecting egg production can also affect egg size e.g Newcastle disease and infectious in laying hens.

SHELL QUALITY

Shell colour is determined by the genetics of the bird and it is mainly a breed and individual bird characteristic and is not related to flavor, nutritional value or the keeping quality of the eggs. Some breeds such as the white leghorn lay white shell eggs. Others such as mode island red, shavers, harco they lay brown shell eggs. The difference, depends on the presence or absence of a pigment secreting mechanism in the uterus or shall gland. The pigment involved is called **PROPORPHYRIN** (from the breakdown of the red blood cell).

SHELL STRENGTH:

MEASURES OF THE SHELL STRENGTH:- They are usually expressed interms of;

1. Shell weight; measure the eight of the shell
2. Shell percentage; measure the weight of the shell divide by the weight of the egg x 100
3. Shell thickness; using a micrometer screw gauge measure the shell (mm). About 0.35mm is good.
4. Specific gravity; Hold the egg and apply pressure, the higher the pressure to break the higher the quality
5. Shell strength; Use, brine solve to find the density 1.07-1.10 is good

All these are measures of shell quality; indirect method of determining shell quality without destroying the egg.

FACTORS AFFECTING SHELL QUALITY

1. **GENETICS OR INHERITANCE:** The ability to produce strong firm shells is inherited. As a result of this selective breeding has lead to the production of good quality shell lines.

2. **NUTRITION:** Several nutritional factors have been reported to be important in shell formation. These include, calcium, phosphorus and vitamin D₃ (chole – calciferot) as a hormone stimulates the absorption of Ca & P in the intestinal mucosa.
3. **AGE OF THE BIRD:** On the average shell quality decreases as the laying hens grow older. As the egg size increases the limited amount of calcium need be spread round the egg.
4. **HEALTH OF THE FLOCK:** Research has shown that infectious bronchitis, Newcastle diseases, largngo trichitis and intestinal coccidiosis cause birds to lay eggs with markedly reduced shell quality. Respiratory diseases outbreak usually result in eggs of poor shell quality which may persist for several months.
5. **Environmental temperature:** High temperature leads to reduced feed intake which results in low intake of Ca, P, & Vitamin D₃ leading to thin shell. Also hyperventilation during hot temperature leads to panting which leads to loss of CO₂, CaCO₃ which will deposited Ca.

Egg processing

Eggs are picked up frequently from the farm and transported by refrigerated trucks to the egg processing and grading station. Eggs are then kept in refrigerated receiving rooms, then washed and dried by a mechanical washer and drier. Eggs then pass through a lash candling machine. After candling, eggs pass through a weighing machine which separates them into various sizes depending on the weight (Jumbo, very large, large, medium, small and peewee). After separation into sizes we have packing into cartoons or trays which can be done manually or automated packing machine. The cartons are then closed and sealed and transported in refrigerated trucks to marketing outlets. In Nigeria we deal mainly with shell eggs but in developed countries there are various production.

Eggs can be broken and sprayed dried to form dried egg product requires 4kg of liquid whole egg to make 1kg of dried egg powder. You need 2.3kg of liquid yolk to make 1 kg of dried yolk powder, 8.1kg of liquid egg white to make 1kg of dried egg white.

Note: Because of the prevalence of salmonella infection of commercial egg, all broken out eggs for use must be pasteurized. Pasteurization involves heating liquid egg to a range of 60 – 62°C (140 – 143°F) and holding for period of 3.5 – 4 minutes.

POULTRY MEAT

There are 3 main sources of poultry meat.

1. Spent layers: The meat is usually very tough.
2. Cockerels (males of the egg type): Have slow rate of growth and poor feed conversion
3. Broilers: These are birds that have been deliberately developed for meat production.

They have a fast rate of growth and good feed conversion. They grow very fast and can reach market weight in about 8 weeks with a feeds conversion of 2 (2kg of feed for 1 kg of meat)

To produce meat from these birds they must be killed and processed. They functions are done in the processing plant. The size of the processing plant is determined by the number of birds processed per hour. To do this you have a division of labour in the poultry industry. CONTRACT GROWER will supply labour and expertise, will supply housing for the birds and veterinary care. The contract grower is paid per live bird produced.

The processing plant will supply the day old chicks to the contract grower, also provide feed to be used, transportation (carrying the birds from the contract grower to the processing plant).

You can have processing plants processing 500,000 to 1 million birds a day.

BROILER PROCESSING PLANT

Plants for dressing poultry vary in size depending on the number of birds processed per hour. Modern processing plants are efficient and continuous line facilities in which the birds are moved from one operation to another one via a monorail.

Live birds are shackled, they are electrically stunned, they are bled, scalded to facilitate feather removal, plucking of feathers, eviscerated, the carcass is inspected by veterinarians, thoroughly washed, chilled, drained of water and packaged.

SLAUGHTER AND BLEEDING

Generally birds that are to be processed are not fed for 12 hours before slaughtering. This is to ensure that their crop is empty which makes for cleaner operation. The bleeding time depends on the efficiency of the cut of the throat the type of bird and whether the bird was electrically stunned or not before slaughter. Bleeding may take 1-3 minutes. Bleeding must be quite complete in order to produce the desirable colour in the final dressed bird.

Scalding

After bleeding the birds are conveyed through a scalding tank scalding loosens the feathers and makes for easier plucking and pin (thin) feather removal (hairy feathers). The higher the temperature the shorter the time required but careful time and temperature control is very important because at higher temperatures there is higher (greater) danger of removal of part of the skin in the defeathering machine. Scalding may be done at 60°C (140°F) in about 45 seconds or more safely with less chance of skin removal 52°C (125°F) in about 2 minutes.

DEFEATHERING

Is done mechanically using a device that has a multiplicity of rotating rubber fingers. This removes all the feathers and the remaining thin feathers are removed by passing the carcass through a flame.

EVisCERATION

This involves cutting the abdominal area to expose the viscera. Evisceration include inspection of the viscera by veterinary agents the lungs to other organs are removed by suction tubes.

WASHING

The washing water can also contain disinfectants to reduce the bacteria load of the carcass and increase the keeping ability.

After inspection birds are thoroughly washed to remove contaminants and reduce bacteria load from the carcass.

CHILLING

The washed birds are now rapidly chilled from about 32°C to 1.7°C within 2 hours to prevent bacteria spoilage and to preserve quality. Chilling is done with ice-slush and the birds absorb a small amount of moisture from the slush which makes them more succulent.

After chilling the carcasses are drained of excess moisture, sized and graded for quality

DRAINAGE OF CARCASS

After chilling the carcass could be in form of cut-out-parts or complete.

PACKING

They are kept in refrigerated trucks and transported to outlets. There could be frozen or refrigerated storage.

OTHER TYPES OF MEAT

MEAT AND MEAT PRODUCTS

Meat and meat products include the skeletal tissue of flesh of cattle, pigs, sheep, goats and organs such as tongues, liver, heart, brain and other edible parts. This represents the red meat of four legged animals. In Nigeria the principal sources of red meat are cattle, sheep, pigs and goats. In the processing of these animals we have some milk product and some by products

e.g of some by product fat, tallow, lard which can be used as energy sources in feed, meat and bone meal, bone meal hides and skin, wool, blood meal (branch chain amino acid valina, leucina, isoleucine, it has low digestibility). In addition to, farm animals yielding meat which is one of our most important source of high quality protein, vitamins, minerals and other nutrients the management of meat yielding animals also contribute to soil fertility and is a means of converting plant roughage material; unsuitable for human consumption into highly acceptable human food.

RED MEAT

The animals are electrically stunned then the throat is cut for proper bleeding. The animals are skinned, after which they are eviscerated and veterinarians to check the viscera of the animals is inspection. After which the carcass is thoroughly washed. The fresh meat is cut into parts. It could be chilled or frozen to keep for a long time.

Meat from various sources represents the muscle of animals and must be graded and inspected for wholesomeness (clean).

Grading is based on such factors as colour and contour of the meat, amount of fat, degree of marbling (fat between the muscle of the meat) of the fat, texture and firmness of the lean. Generally the best cut of meat have some fat and the fat is well marbled throughout the lean. This results in greater tenderness and better flavor. The grades of meat have very little relationship to the nutritional value of the cut and grading is usually done at the place of slaughter.

The purpose of meat inspection is to safeguard health by ensuring a clean, wholesome, disease free meat supply that is without adulteration if animals are diseased the meat can carry a wide variety of organisms pathogenic to man. This may include species capable of causing tuberculosis, brucellosis, anthrax, trichinosis and salmonellosis. These are some of the

animal diseases that can be transmitted to man and this is why meat inspection must be made by trained veterinarians at places of animal slaughter and at meat processing facilities.

As a rule animals to be slaughtered for meat should not be agitated. When agitated the glycogen is converted to energy which is supposed to be stored as lactic acid.

SLAUGHTERING

As a rule animals are rendered insensitive to pain before being hoisted by their hind legs and stuck in the neck for bleeding. After stunning, hoisting and bleeding every component is utilized including the viscera, blood, hide and the carcass. The skinned, washed and eviscerated carcass is then moved by manual into a chilled room whereas the deepest part of the meat reaches 1.7°C (35°F) in about 36 hours. This prevents rapid bacterial spoilage.

CONSIDERATION FOR QUALITY MEAT

Disease free animals

No agitation of the animal before slaughter

Thorough washing

Rapid chilling

Resting of animals before slaughter is another practice to help delay bacterial spoilage. Animals store glycogen in their muscle as a source of reserve energy. After the animal is killed, this glycogen is converted under anaerobic condition in the muscle into lactic acid. This gives the muscle increased acidity which acts as a mild preservative but if animals are excited or exercised before slaughter the glycogen is largely consumed and there is very little left to be converted to lactic acid in the postmortem tissue. Such meat can spoil more quickly this is one reason for the common practice of resting animals before slaughter.

Muscle is composed of bundles of hair like muscle fibres. These protein muscle fibres are held together by proteinaceous connective tissues form a tendon which in turn to a bone (tendons connect muscle to bone, while ligament connect bone to bone). The muscle fibres

themselves are elongated cells that contain many smaller fibrils. A major protein of muscle fibre is myosin. The connective tissue contains to protein called collagen and elastin.

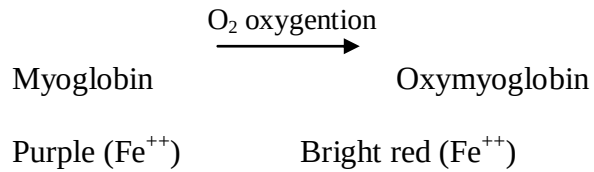
Collagen on heating in the presence of moisture dissolves and yield gelatin. Elastic is tougher and is a constituent of the ligament. There is a relationship between muscle structures and meat tenderness. Thinner muscle fibres are more tender than thicker muscle fibres and thinner muscle fibres are more common in-young animals. Thus meat from younger animals are more tender than meat from older animals.

On cooking muscle fibres will contract and cooking also meet the fat and dissolve the collagen into soluble gelatin and the overall effect is increased tenderness of the meat. The composition of meat from various animals vary.

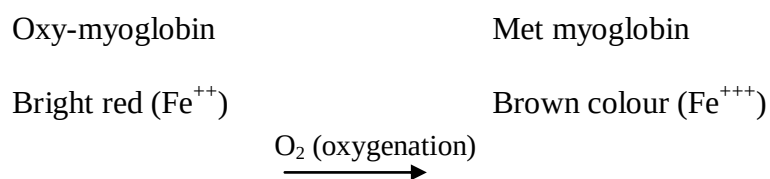
Chemical components	Chickens	Beef	Mutton	Park	Milk
Protein	20	18	16	12	3.5
Fat	13	21	28	45	3.5
Water	66	60	56	42	88
Carbohydrate	-	-	-	-	-
Ash	1	1	1	0.6	0.7

MEAT COLOUR CHANGES ASSOCIATED WITH TIME

Meat colour is very important from the stand point of marketing. One of the principal objective of curing is to stabilize the red colour and this require some understanding of the pigment of the meat and their colour changes. This is most important because the consumer attaches great significance to red meat colour. The chief muscle pigment is a protein called myoglobin and it has a publish colour. When it is exposed to oxygen it ncomes oxymyoglobin which has a bright red colour.



Thus, when fresh meat is first cut it is publish red in colour but its surface quickly become bright red upon exposure to air. The desirable bright red of oxy-myoglobin on exposure to air is not entirely stable and on prolong exposure and excessive oxidation it can shift to met myoglobin which has a brownish colour.



When fresh meat is cooked these protein pigments are denatured and the act of cooking produces a brown colour. Well done meat is more denatured and is more brownish but meat cured with nitrate and nitrates are red and remain red through cooking. These pigment shift some of which are reversible are affected by oxygen acidity of the meat and the combination of these determine which pigment will dominate without the normal pigment shift the colour of the meat does not indicate wholesomeness or nutritional value. However, the red colour is an important positive sales influences. For this reason packaging materials for meat re designed to protect meat colour largely by controlling diffusion of oxygen.

In the case of fresh meat packages are used which allow air to penetrate and keep myoglobin in the bright red oxymyoglobin form.

PRINCIPLES OF MEAT PRESERVATION

Meat is a very perishable products and condition that favour the growth of microorganisms will surely reduce the shelf life of meat products. Meat preservation methods are based on the following principles.

1. Reducing the activity and growth of microorganisms through the use of low temperature e.g freezing, chilling and refrigeration of meat.
2. Reducing the activity and growth of microorganisms through the reduction of water content of the meat cooking drying, roasting, frying etc.
3. Reducing the activity and growth of microorganisms through the use of chemicals as preservatives e.g salting, curing etc.

Some of the preservation methods include salting and curing, freezing drying, chilling etc

The most important aspect of meat preservation is that we start with a clean uncontaminated wholesome product with low bacterial load. The low bacterial load on the carcass is achieved by not agitating the animal before slaughter, thorough washing of the carcass with water during processing and by chilling the carcass rapidly to achieve a temperature below 35°F between a few hours of slaughter such meat will now respond very well to preservation method such as salting, curing, cooking, drying and freezing.

CURING

Curing refers to the modification of the meat that affects the preservation of colour flavor, tenderness due to added curing ingredients. Curing is designed to grossly alter the nature of the meat and produce distinct products such as smoke and salted bacon, ham, corn beef, salamis, biltong. Curing is a method of meat preservation and it was done before refrigeration was developed. In less developed countries without modern preservation facilities, the prime objective of curing is preservation.

In the developed countries, the prime process of curing is to produce unique flavoured meat products and a special purpose is to preserve the red colour of meat. Thus, cured corn beef, well cooked remains red in colour while beef that have not been cured turns brown on cooking. The principal ingredients used for curing meat are;

1. Common salt(sodium chloride): It is a mild preservative and adds to flavor

2. Sodium nitrate (Na NO_3) and sodium nitrite (Na NO_2): They are red colour preservatives.
3. Sugar: It helps to stabilize the colour and also add to flavor
4. Spices: They make add to the flavor of the meat.

The cure may be applied to meat in the dry form by rubbing the surface of the meat or a solution of the cure can be made and the meat soaked in the cure or the cure can be injected into the meat.

Smoking

Following curing processed meat may be smoked. Smoking adds as a mild preservation and it is used mostly because of the flavor contribution. Smoking is done in large smoke houses by hanging the meat over burning hard wood logs or wood chips and the smoke wood is maintained at about 52°C (125°F). Smoking may take about 18-24 hours. The meat may be cooked before smoking or may be cooked after smoking. When smoking pork you must have about 58°C or higher to ensure the destruction of trichinosis parasite.

FRYING

Cooked meat keeps better than raw meat especially if the meat is also fried after cooking. Cooking can make meat more or less tender than the original raw meat. When meat is cooked there are three.

- (1) The fat melts and contributes to the tenderness of the meat
- (2) The connective tissue collagen also dissolves in the hot liquid and becomes the soft gelatin. Then the muscle fibres also separate.

DRYING

Meat drying involves cutting the raw meat into pieces, sorting the meat and drying it in the sun for several days until you have a very dry compact product, which may be cooked afterwards before consumption. Sometimes the raw meat is first cooked and salted before drying

in the sun to give a very dry product which can be stored for long without spoilage. Preservation is based on the reduced water content of the product, as well as the activities of ultraviolet rays of the sun on the microbial load of the meat. This is the basis of such meat as kilishi or other dry meat method.

MILK AND MILK PRODUCTS

Milk is the normal secretion of the mammary glands of all mammals. Its purpose in nature is to nourish the young of particular specie of animal producing the milk. The nutritional needs of different species vary and so it is not surprising that the milk for different mammal differs in composition.

SOURCES OF MILK

Milk for food apart from human milk can be gotten from large domesticated animals like

- the cow,
- the buffalo,
- the camel,
- the sheep
- and the goat.

All these animals are herbivores.

The cow is the principal source of milk for human consumption throughout the world but there are some few exceptions.

1. Most milk in India is obtained from the water buffalo.
2. In southern European the milk is from sheep and goat.
3. In the Africa area the milk is from the reindeer.

Food value of milk

Milk is a food of high nutritional value.

Its importance is based on the following scientific facts:

- It is an exceptionally good source of protein which is of high biological value in promoting the growth of children.

- It is good source of calcium in the diet and consequently encourages sound bone and tooth development.
- It contains a useful miscellany of vitamin A, thiamine (Vit B1), riboflavin, pyridoxine (Vit B6), biotin, niacin, pantothenic acid, vitamin D, etc.
- The most important component is butterfat, which gives milk its special creamy taste and color. Butterfat is used to make many different products such as cream, butter, butter oil and ghee.

% NUTRIENT

Species	Total solid	Fat	Crude protein	Casein	Lactose	Ash
Cow	12.6	3.8	3.35	2.78	4.75	0.7
Goat	13.18	4.24	3.7	2.8	4.51	0.78
Sheep	17.0	5.3	6.3	4.6	4.6	0.80
Water buffalo	16.77	7.45	3.78	3.0	4.88	0.78
Human milk	12.57	3.75	1.63	-	6.98	0.21

The milk from the udder of the cow is known as the parent milk can be separated into two parts;

(A) Cream (fat)

(B) Skim milk (protein and carbohydrate).

This can be sold as products on their own or they can be further processed into butter, cheese, ice cream, yoghurt, condensed milk, evaporated milk, dry milk, and other milk products.

Furthermore, the parent milk may be modified by condensing, drying, flavouring or fortifying.

The principal constituents of milk include;

1. Fat
2. Protein (casein)
3. Milk sugar (lactose)
4. Minerals
5. Water

The solid in the milk can be classified into two

1. Fat milk
2. Non-fat solid

The grade of the milk is dependent on the fat. The composition of milk varies depending on the species of the animal and the bulk of milk consumption of human comes from the cow. The milk from cow varies in composition depending upon many factors;

1. The breed of the cow
2. The individuality of the animal
3. The age of the animal
4. The stage of lactation
5. The season of the year
6. The feed of the cow
7. Time of milking
8. The period of time between milking
9. Drugs and medication given to the animal
10. The health of condition of the animal

The total solid of any cow milk is about 12.9%, the rest water. That is why dairy production requires a lot of water. The total solid non-fat is about 9% (protein, CHO some minerals). The market price of milk purchase in bulk generally is based on the fat content.

There are some production modes from milk (butter, cheese, evaporated milk, evaporated cream, dry milk, yoghurt). The exotic breeds of dairy animals are;

1. Jersey
2. Holstein Frisian
3. Guernsey
4. Brown swiss
5. Agrshire

The Holstein Frisian generally produce the highest quantity of milk but the Guernsey and Jersey produce milk with the highest fat content (5%)

MILK PROCESSING

The word **DAIRY** entails Milk and its products.

What is milk processing?

Milk processing includes all simple methods used to handle milk with a view to

- ☐ Improve the quality of milk and milk products into high value, concentrated and easily transportable dairy products such as butter, cheese and yogurt.
- ☐ Preserve milk for days, weeks and months.
- ☐ Reduce food-borne illness.

Importance of milk processing

- ☐ Milk processing is an important source of employment and income.
- ☐ It is a provider of high quality human Nutrition in many developing countries.
- ☐ Good knowledge on milk production and processing enhances better control and prevention of milk wastage.
- ☐ Selling processed milk products is more profitable than selling fresh milk.

POSSIBILITIES OF INFECTION

The rich nutritional composition of milk consisting of fat, protein, lactose (milk sugar), mineral salts and vitamins, gives rise to its main disadvantage from the point of view of a dairy technologist. This is that milk is readily infected by microorganisms.

DISEASES THAT CAN BE TRANSMITTED VIA MILK

	IN ANIMAL	IN MAN	MODE OF INFECTION
Mycobacterium tuberculosis	T.B. in cattle	T.B. in man	From infected udder
Brucella abortus	Abortion in cattle	Undulant fever	From infected udder
Staphylococcus aureus	Mastitis	Food poisoning	From infected udder and personnel
Streptococcus pyogenes	Mastitis	Scarlet fever Septic sore throat	Ditto
Rickettsia		Acute fever	From infected udder
Corynebacterium diphtheria	Ulceration of teeth	Diphtheria	Utensils and personnel
Salmonella typhi		Typhoid fever	Ditto
Salmonella paratyphi		Paratyphoid fever	Ditto
Salmonella Dublin	Enteritis	Gastro-enteritis	Ditto
Salmonella typhimurium	Enteritis	Gastro-enteritis	Ditto
Shigella dysenteriae		Dysentery	ditto

Micro-organisms derived from milk:

- The cow, or other producing animals
- The equipment and utensils in which it is processed.
- The people who handle it, and
- The atmosphere.

OBJECTIVES OF MILK PROCESSING

- ❖ To destroy pathogens
- ❖ To improve the keeping quality of the milk.
- ❖ To inactivate spoilage enzymes e.g. lipase
- ❖ To help produce convenience items e.g. instant dried milk, cheese.
- ❖ To transform or convert milk into products preserved for human consumption e.g. yoghurt and cheese.
- ❖ To improve the nutritional characteristics of milk.
- ❖ To facilitate the handling of milk through reduction in weight and bulk.
- ❖ To improve the appearance, aroma and flavour of milk.

PRODUCTS OF MILK PROCESSING

Milk can be processed into these following:

1. Market milk (pasteurized standardized milk)
2. Market/filled milk
3. Skim milk
4. Homogenized milk

5. Evaporated milk
6. Sweetened condensed whole/skim milk
7. Instant dried whole/skim milk
8. Ice milk
9. Yoghurt
10. Acidophilus milk
11. Butter
12. Cheese – (soft or cottage, semi-hard, and hard)

Small scale milk processing

Small scale milk processing can best be demonstrated as milk processing in which a small house-holder can utilize available kitchen utensils to process milk without destroying the essential nutrients in it.

YOGHURT PRODUCTION

INTRODUCTION

The Word yoghurt” or “yogurt “reflected to the Turkish word” Jau-urt,” which means thick milk. It is sour milk known for many centuries. The indigenous yoghurts are mainly manufactured in the middle-East and in other countries around the Eastern part of the Mediterranean. They are known as mast (Iran), leban or laban Iraq, Lebanon, Egypt, Zebedee (Sudan), Delhi is a yoghurt-like product of Indian. In many cases, yoghurt is actually milk product produced by bacterial fermentation of milk.

Worldwide, cow’s milk, the protein of which is mainly Casein is most commonly used to make yoghurt. Milk from water buffalo, goats, ewe, mare, yaks, however is also used to produce yoghurt in various part of the World. The name ‘yoghurt ‘ should be preserved for products manufactured by cultures consisting exclusively of Streptococcus Thermophiles and Lactobacillus bulgaricus.

Yoghurt can be produced in small scale basis by the methods stated below:

MILK COLLECTION



Adulteration test



- (i) Dipping a finger in a small cup of milk sample. Normal milk has a retaining firm in the finger.



(ii) Use of lactometer (1.028-1.032) at 20°C.

Clot-on boiling test.



(iii) If you put a sample on fire and it boils without clotting then the milk is good.

Measuring of milk



Measure the milk usually in litres to ascertain the quantity to be processed.

SIEVING

Filter cloth is used to remove sediments and reduce microbial load.



PASTEURIZATION

Methods of Pasteurization

- ❖ Batch pasteurization: this is heating milk at a temperature of 63°C for at least 30 minutes. This is suitable for small-scale producers and farmer cooperatives.

- ❖ High temperature short time (HTST) pasteurization: This is heating milk at a temperature of 72°C for at least 15 seconds. This is suitable for processing large quantities of milk, e.g. more than 250 liters at a time.

Importance of Pasteurization

- To destroy pathogens and organisms in the milk.
- To provide a cleaner medium in which the desired organism can be established.
- To remove air from milk.
- It denatures the whey proteins which increases the viscosity of the product.



COOLING

The heated milk is immediately immersed in a basin of cold water to bring the milk to a temperature of 40°C – 45°C to:





Benefits of fast cooling

- Subject thermophilic organisms in milk to thermal shock resulting in autolysis.
- Provide a convenient environment for the inoculation of starter organisms.

PACKAGING OF PASTEURISED MILK

The pasteurized milk is now packaged in sterile containers for storage or table.

INOCULATION

YOGHURT STARTER CULTURE

A yoghurt culture consist of two bacteria both belong to the family of the Lato bacillaceae and both are thermophilic:

- a) *Lactobacillus bulgaricus*.
This is a gram + bacillus responsible for acid production as well as for the production of the typical yoghurt aroma (ethanol).

b) *Streptococcus thermophiles*.

This is a gram + coccus. A fast grower in milk. It produces lactic acid and a little formic acid which stimulate the development *Lactobacillus bulgaricus*. The *Lactobacillus*. The *Lactobacillus* produces amino-acid that stimulate the growth of the cocci.

Streptococcus. Thermophile =====> Produces ===> Formic acid

Stimulate

Amino acid <==== *bulgaricus* <====> *L. bulgaricus*

Milk protein.

Symbiosis = *L. bulgaricus* <====> *Streptococcus. thermophile*

Mix pasteurized milk with yoghurt obtained from a reliable source e.g. NAPRI yoghurt. Then stir properly.

Inoculums should not be more than 2%.





Incubation

After pasteurization of milk,

- (a) Pour warm water of temperature 45°C into a big warmer or basin and cover. The aim is to sterilize the warmer or basin and to warm it up.
- (b) Ensure that all containers and spoons that will come into contact with the milk are sterilized in hot water.
- (c) Transfer the inoculated milk into a Basin of warm water of temperature 45°C
- (d) Cover the basin or warmer and allow milk to coagulate within 2-3 hours. If the container is not food flask, lag properly to prevent heat loss.



Cooling

After coagulation, allow yoghurt to cool in a refrigerator or freezer of about 10°C or less.



Stirring

Stir the already formed yoghurt properly. This will help to homogenize (break it into equal sizes) it



Addition of additives:

Preservative and other additives like flavors, colorings, stabilizers etc. can be added.



Package the yoghurt

This is dividing the yoghurt into smaller units for easy sharing and marketing.



Advantages of yoghurt

- (i) It has improved keeping quality.
- (ii) It has improved nutritive value.
- (iii) It is antibiosis with low (pH 4.6, 1- 1.2% lactic acid) and contains antibiotics (i.e. nisin).
- (iv) It has therapeutic/prophylactic properties on intestinal ailments and has antitumor activity.
- (v) It helps in the alleviation of lactose intolerance.
- (vi) It enhances absorption of calcium.
- (vii) It acts as a complete meal when taken with mashed Fura (Millet Balls) or with corn flakes.
- (viii) It is a source of income and a provider of labour.

Disadvantages of yoghurt

- i. Overweight from too much consumption
- ii. Lactose intolerance